Kepler's laws

(of planetary motion)

Kep's laws apply, in general, to orbital situations such as a moon orbiting its planet.
It is NOT exclusive to the orbiting w.r.t. the Sun.

1. Planets move in circular orbits with the Sun at one focus.
2. The line connecting a planet to the Sun sweeps out equal areas in equal times.
3. The square of the periodic orbital time T of a planet is directly proportional to the cube of the orbital radius r: $T^2 \propto r^3$.

Remember:

1. ... move ...
2. ... line ...
3. ... square of the periodic orbital time ...
   MLSqr

TODO: Add link with topic A2

The gravitational force of a planet provides the **centripetal force** for a satellite in a circular orbit. The force between the satellite and planet is at right angles to the direction of motion of the satellite.

Hence,

$$\begin{aligned}
\text{Let eqn 1 be } \ g &= G\frac{M}{r^2} \\
\text{Let eqn 2 be } \ a &= \frac{v^2}{r} \\
\text{centripetal acceleration} &= \text{grav. field strength} \\
a &= g \\
\therefore \frac{v^2}{r} &= G\frac{M}{r^2} \\
v &= \sqrt{\frac{GM}{r}}
\end{aligned}$$

M is the mass of the planet / the spherical body that the satellite is orbiting.